

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Course Code/ Name: DSA0601 /Data Handling and Visualization

Slot C | CO1 – AT3 (R Only)

Q1: Color Scales for Multi-Variable Data [R]

City	Temp	Humidity	Rainfall	AQI
Delhi	44	22	6	365
Mumbai	35	84	280	110
Chennai	38	78	220	125
Kolkata	40	72	190	180
Hyderabad	39	72	85	160
Bangalore	31	66	120	72
Pune	33	58	95	84
Ahmedabad	43	26	2	290
Jaipur	45	18	1	320
Lucknow	39	42	35	240
Bhopal	38	36	48	175
Patna	40	45	72	220
Goa	32	89	360	58
Kochi	30	91	410	52
Nagpur	41	30	12	270
Indore	37	34	20	190
Surat	36	76	180	145
Visakhapatnam	34	82	240	98
Coimbatore	29	68	130	60
Mysuru	28	64	145	55

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents Temperature, Air Quality Index (AQI), Rainfall, and Humidity using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

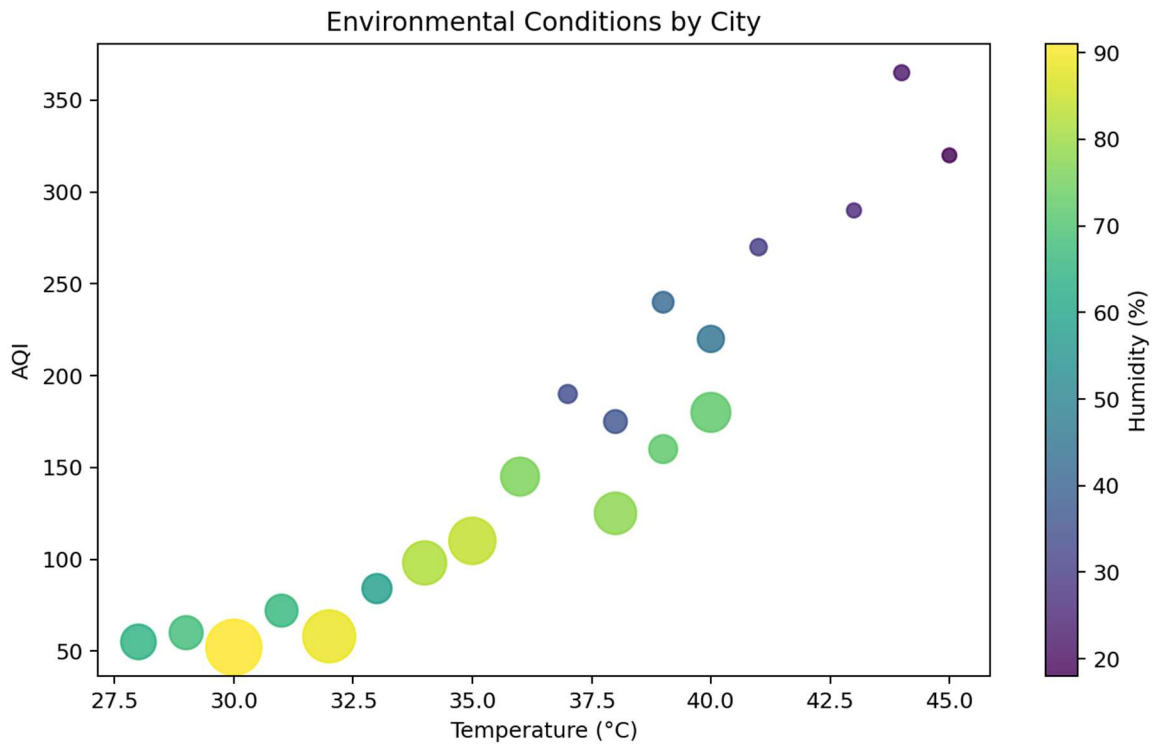
Answer / Design

Use a bubble scatter plot: Temperature → x-axis; AQI → y-axis; Rainfall → bubble size; Humidity → colour. A sequential Viridis colour scale is suitable because humidity is ordered numerical data and Viridis is perceptually uniform and generally colour-blind friendly.

```
library(ggplot2)
```

```
ggplot(city_data, aes(x=Temp, y=AQI, size=Rainfall, color=Humidity)) +  
  geom_point(alpha=0.8) +  
  scale_color_viridis_c() +  
  labs(title="Environmental Conditions by City",  
        x="Temperature (°C)", y="Air Quality Index (AQI)",  
        size="Rainfall", color="Humidity (%)") +  
  theme_minimal()
```

OUTPUT:



Q2: Integrated / Capstone Design [R]

<i>Company</i>	<i>Sector</i>	<i>Revenue</i>	<i>ProfitMgn</i>	<i>Employees</i>	<i>MarketCap</i>
C1	IT	420	8.2	1500	8500
C2	Banking	580	15.6	3200	12000
C3	Healthcare	720	11.4	2100	9800
C4	Retail	850	9.3	4200	7600
C5	Energy	980	18.5	2800	14500
C6	IT	1150	20.4	5100	18200
C7	Banking	1320	17.8	6800	21500
C8	Healthcare	1480	14.2	3900	16800
C9	Retail	1650	10.5	7200	13200
C10	Energy	1820	22.1	4500	24500
C11	IT	2050	24.5	8100	28500
C12	Banking	2280	19.8	9500	32500
C13	Healthcare	2550	16.7	6200	24800
C14	Retail	2840	13.4	10300	19200
C15	Energy	3150	26.2	7200	38200
C16	IT	3480	28.1	11800	42500
C17	Banking	3860	21.5	13200	46800
C18	Healthcare	4250	18.3	8600	35600
C19	Retail	4680	15.2	14500	29800
C20	Energy	5200	30.4	16000	52000

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry/Sector. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

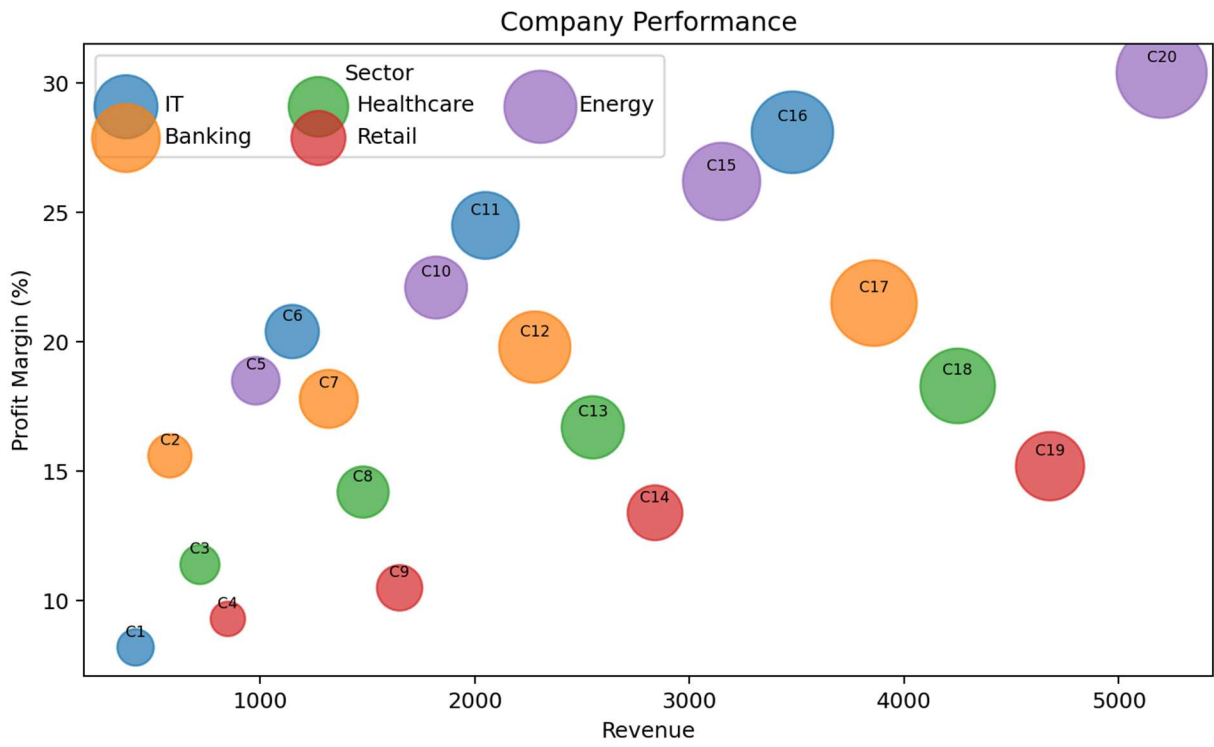
Answer / Design

Use an integrated bubble scatter plot: Revenue → x-axis; Profit Margin → y-axis; Market Capitalization → bubble size; Sector → colour; Company → label. Employee Strength can be included as a tooltip in an interactive version, or as an additional annotation when required.

```
library(ggplot2)

ggplot(finance_data,
       aes(x=Revenue, y=ProfitMgn, size=MarketCap, color=Sector)) +
  geom_point(alpha=0.75) +
  geom_text(aes(label=Company), vjust=-1, size=3) +
  labs(title="Company Performance",
       x="Revenue", y="Profit Margin (%)",
       size="Market Cap", color="Sector") +
  theme_minimal()
```

OUTPUT:



Q3: Multi-Aesthetic Design [R]

Branch	Region	Category	Sales	Profit	Customers
B1	North	Electronics	450000	45000	1200
B2	South	Furniture	620000	95000	2500
B3	East	Clothing	180000	-12000	980
B4	West	Food	920000	185000	4200
B5	North	Electronics	350000	25000	1800
B6	South	Furniture	720000	120000	3000
B7	East	Clothing	250000	18000	1250
B8	West	Food	1080000	225000	5200
B9	North	Electronics	560000	72000	2100
B10	South	Furniture	670000	98000	3300
B11	East	Clothing	195000	-8000	1100
B12	West	Food	980000	205000	4700
B13	North	Electronics	430000	51000	1750
B14	South	Furniture	760000	132000	3600
B15	East	Clothing	280000	24000	1450
B16	West	Food	1150000	250000	5600
B17	North	Electronics	510000	69000	2200
B18	South	Furniture	690000	115000	3900
B19	East	Clothing	220000	15000	1600
B20	West	Food	1250000	285000	6100

A multinational retail company has collected annual sales information from its branches across different regions. Design an effective visualization that simultaneously represents Sales, Profit, Number of Customers, Product Category, and Region. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.

Answer / Design

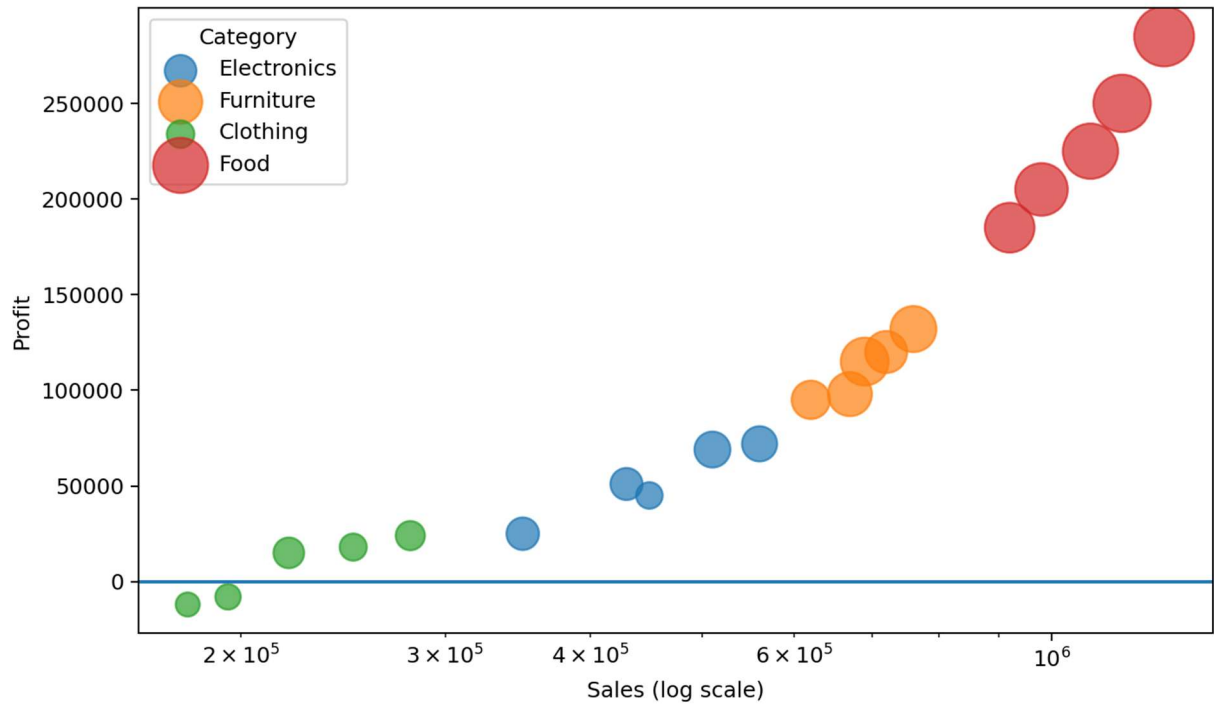
Use a multi-aesthetic scatter plot: Sales → x-axis; Profit → y-axis; Customers → point size; Product Category → colour; Region → shape. A logarithmic Sales scale helps when values span a wide range. A horizontal zero line separates positive and negative profit.

```
library(ggplot2)
```

```
ggplot(branch_data,  
  aes(x=Sales, y=Profit, size=Customers,  
      color=Category, shape=Region)) +  
  geom_hline(yintercept=0) +  
  geom_point(alpha=0.75) +  
  scale_x_log10() +  
  labs(title="Branch Sales, Profit and Customers",  
       x="Sales (log scale)", y="Profit",  
       size="Customers", color="Product Category",  
       shape="Region") +  
  theme_minimal()
```

OUTPUT:

Branch Sales, Profit and Customers



Q4: Cartesian vs. Polar Coordinates [R]

<i>Month</i>	<i>Coal</i>	<i>Gas</i>	<i>Solar</i>	<i>Wind</i>	<i>Hydro</i>
<i>Jan</i>	620	310	45	80	140
<i>Feb</i>	580	295	60	95	150
<i>Mar</i>	590	305	90	120	160
<i>Apr</i>	605	320	130	150	170
<i>May</i>	640	340	180	170	180
<i>Jun</i>	670	355	230	185	195
<i>Jul</i>	690	360	250	195	205
<i>Aug</i>	720	370	245	190	210
<i>Sep</i>	700	365	210	175	200
<i>Oct</i>	680	350	160	150	185
<i>Nov</i>	650	335	90	110	165
<i>Dec</i>	630	320	50	90	150

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using Cartesian Coordinates and Polar Coordinates. Compare both visualizations and determine which coordinate system provides better analytical insight. Explain your design decisions.

A. Cartesian Coordinates

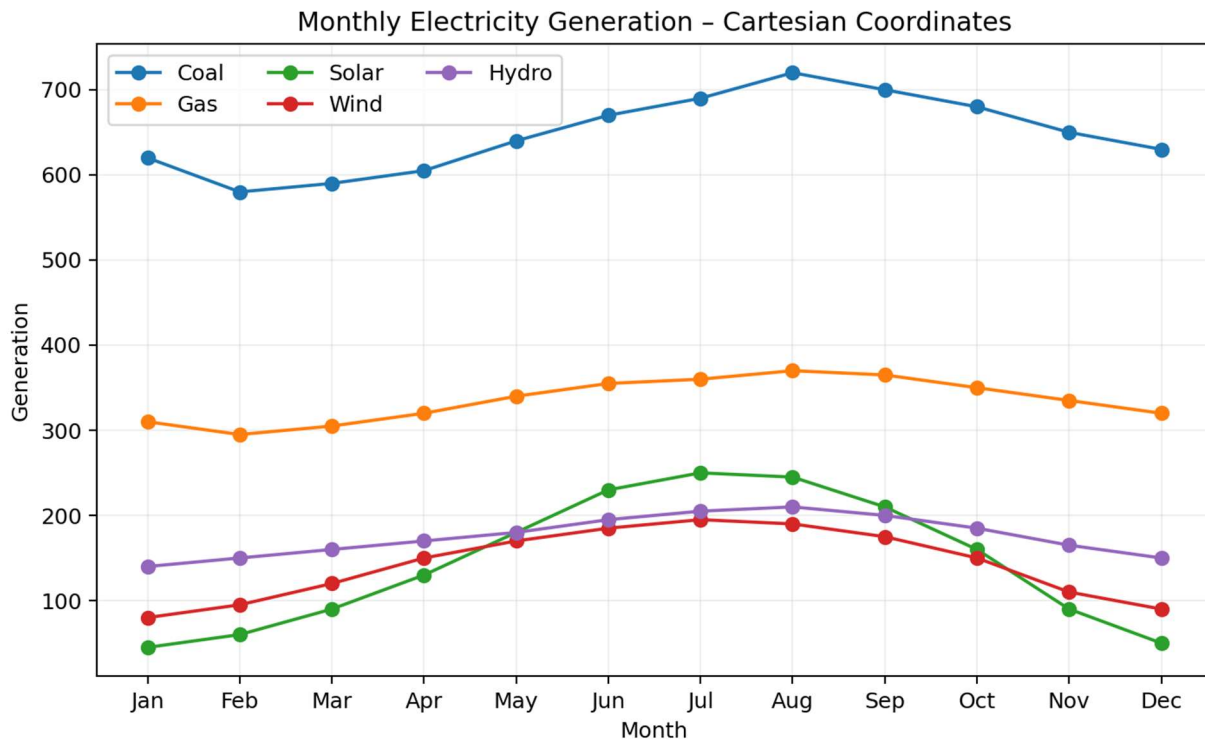
```
library(ggplot2)
```

```
library(tidyr)
```

```
energy_long <- pivot_longer(energy_data, cols=Coal:Hydro,  
                             names_to="Source", values_to="Generation")
```

```
ggplot(energy_long,  
       aes(x=Month, y=Generation, color=Source, group=Source)) +  
  geom_line() + geom_point() +  
  labs(title="Monthly Electricity Generation – Cartesian Coordinates",  
       x="Month", y="Generation", color="Energy Source") +  
  theme_minimal()
```

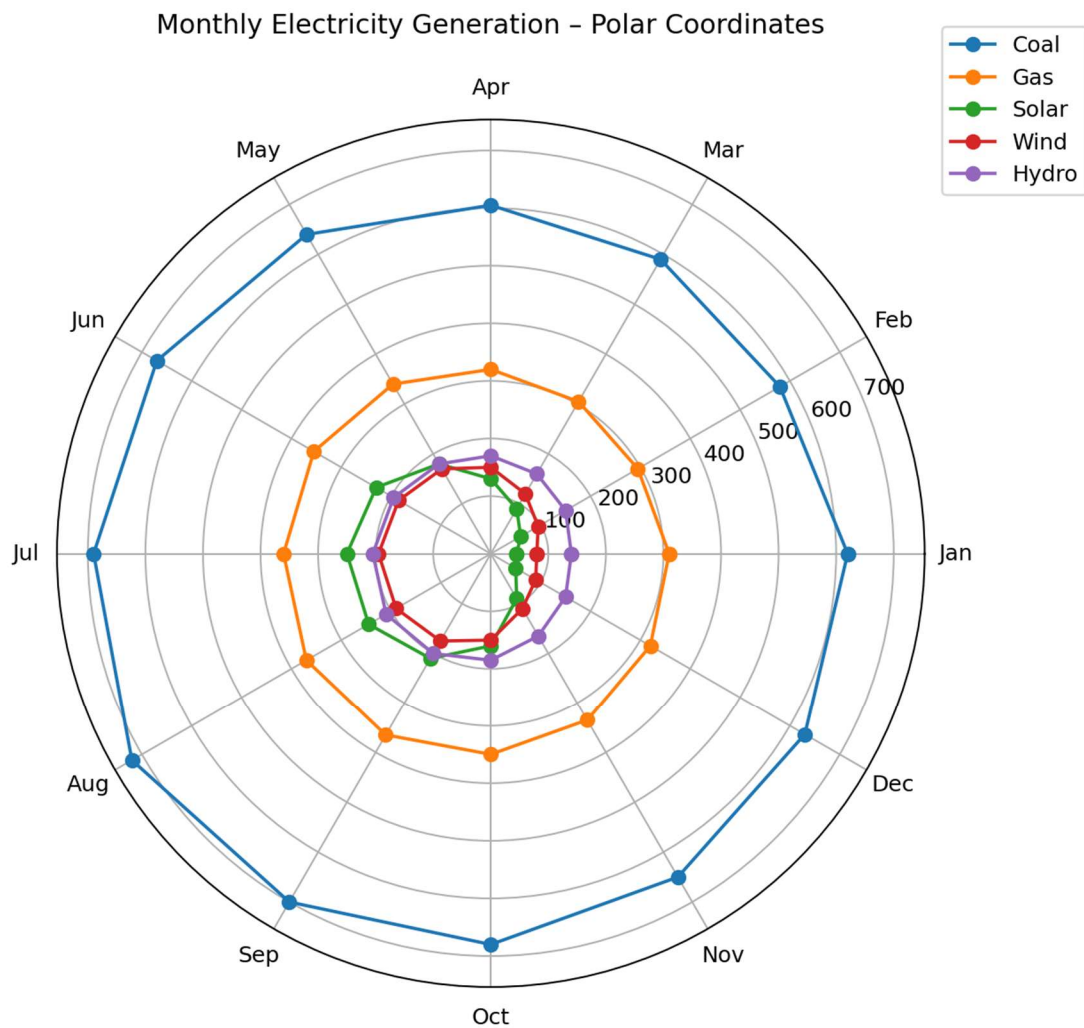
OUTPUT:



B. Polar Coordinates

```
ggplot(energy_long,
  aes(x=Month, y=Generation, color=Source, group=Source)) +
  geom_line() + geom_point() +
  coord_polar() +
  labs(title="Monthly Electricity Generation – Polar Coordinates",
    x="Month", y="Generation", color="Energy Source") +
  theme_minimal()
```

OUTPUT:



Comparison / Conclusion

Cartesian coordinates are better for precise month-to-month comparison because values are positioned on linear axes and are easier to compare quantitatively. Polar coordinates emphasize the cyclical nature of the year and make the December-to-January relationship visually continuous, but radial distances are harder to compare precisely. Therefore, for analytical comparison of electricity generation, the Cartesian line chart provides clearer quantitative insight; the polar chart is useful as a complementary seasonal view.

Quick Design Summary

Question	Visualization	Main Aesthetic Mappings
Q1	Bubble scatter	Temp=x, AQI=y, Rainfall=size, Humidity=color
Q2	Integrated bubble scatter	Revenue=x, Profit Margin=y, MarketCap=size, Sector=color, Company=label
Q3	Multi-aesthetic scatter	Sales=x, Profit=y, Customers=size, Category=color, Region=shape; log Sales
Q4	Cartesian + Polar line charts	Month/time mapped to x or angle; Generation to y or radius; Source=color